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* Back of The Envelope Estimation :-

Q:- Etimate your system capacity

Performance requirements using "Back-of-the Envelope Estimation"

Estimates + Thought experiments + common performance no.

→ will be able to choose which design will meet your requirements.

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must know
- ★ Good sense of 'scalability Basics'
- ★ following concept should be aware :-

→ Power of 2

→ Latency numbers :- किसी Operation को होने में कितना time लगता।

→ Availability numbers

★ power of 2 :-

power	Approximate Value	Full Name	Short Name
10	1 thousand	1 kilobyte	1 KB
20	1 Million	1 Megabyte	1 MB
30	1 Billion	1 Gigabyte	1 GB
40	1 Trillion	1 Terabyte	1 TB
50	1 Quadrillion	1 petabyte	1 PB

★ Latency numbers :-

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Operation name	Time
L1 cache reference	0.5 ns
L2 cache reference	7 ns
Main memory reference	100 ns
Compress 1K bytes with zippy	10,000 ns = 10 μs (micro second)
Send 2K bytes over 1 Gbps network	20,000 ns = 20 μs
read 1 MB from sequentially from memory	2,50,000 ns = 250 μs
Round trip within the same datacenter	500,000 ns = 500 μs
Disk seek	10,000,000 ns = 10 ms (milli second)
Read 1MB sequentially from the network	10,000,000 ns = 10 ms
Read 1MB sequentially from disk	30,000,000 ns = 30 ms
Send packet CA(California) → Netherlands	150,000,000 ns = 150 ms

→ L1 cache reference से 0.5 ns लगते है, data fetch करने में।

CA

California —send→ Netherlands

Netherlands —recieve→ California

⇓

Conclusion :-

→ Memory is faster than disk.
→ Avoid disk seeks if possible.
→ Simple compression algorithms are fast.
→ Compress data before sending if possible
→ Data centers are far located → time taking.

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★ **Availability Numbers :-** कितने time के लिए आपका service available है।

- High Availability = System continuously operational for long.

- Measure in %

  100 % = 0 down time

  Mostly service = 99 % to 100 %

- SLA (Service Level Agreement) :-

  → Agreement b/w service provider and consumer/ customer.

  → कितना available रहेगा (uptime)

→ Ex :- Amazon, Google, Microsoft ⇒ cloud services = 99.9 % ↑

Availability %	downtime per day	downtime per year
99 %	14.40 minutes	3.65 days
99.9 %	1.44 minutes	8.77 hours
99.99 %	8.64 minutes	52.60 minutes
99.999 %	864.00 milliseconds	5.26 minutes
99.9999 %	86.40 milliseconds	31.56 seconds